Sagar Kriplani

Mechanical Design Engineer - Custom Machinery & Automation (m/f/d) — M.Sc. Scientific Instrumentation

Goethestraße 1, 89518 Heidenheim an der Brenz, Germany +49 157 582 90324 sagar.kriplani@outlook.com
linkedin.com/in/sagar-kriplani **Work Authorization:** EU Blue Card (Unrestricted German Work Permit, Available Immediately)

EXECUTIVE SUMMARY

Results-driven Mechanical Design & Development Engineer (M.Sc., Grade 1.8) with over 3 years of hands-on experience in production-ready 3D CAD modeling (Autodesk Inventor 2024, PTC Creo, Siemens NX, SolidWorks), thermomechanical & structural FEA simulation (ANSYS Workbench), and precision system integration. Proven track record managing end-to-end design cycles within custom machinery (Sondermaschinenbau) and semiconductor equipment (8+ machine platforms, 50% energy savings, 70% component cost reduction). Specialized in DFM/DFA, Trumpf sheet metal unfolding, GD&T tolerance analysis (DIN EN ISO 1101), and Sondermaschinenbau, Automatisierungsplattformen, DFM/DFA, Trumpf-Blechabwicklung, GD&T DIN ISO 1101. Business fluent in English (C1) and professional German (B2).

PROFESSIONAL EXPERIENCE

Mechanical Design & Development Engineer — NeSt GmbH Sondermaschinenbau & Anlagenbau, Herbrechtingen, Germany 09/2025 – 09/2026

- **Platform Design & Production Release:** Directed full concept-to-release design cycles for over 8 custom machine platforms, generating ~200 detailed manufacturing drawings per month compliant with DIN/ISO standards.
- **DFM/DFA & Sheet Metal Enclosures:** Engineered modular machine frames, precision mechanics, and sheet metal enclosures (Trumpf unfolding) strictly applying Design for Manufacturability/Assembly principles.
- **FEA Simulation & Tolerance Stacks:** Validated static load integrity and thermal stress behavior using ANSYS Workbench; performed 1D/2D GD&T tolerance stack calculations (DIN EN ISO 1101) to eliminate assembly interferences.
- **Cost & Weight Optimization:** Achieved up to 70% cost reduction and 60% weight savings through component modularization, material optimization, and FMEA-guided design reviews.
- **PDM & ERP Integration:** Managed revision-controlled assembly trees and Bills of Materials (BOM) in PDM and SAGE ERP; coordinated technical interfaces across internal production, assembly technicians, and external suppliers.

Working Student (Mechanical Design - Specialized Equipment) — Brooks Automation (Deutschland) GmbH, Steißlingen, Germany 02/2024 – 06/2024

- **Semiconductor Vacuum Equipment:** Designed high-precision vacuum chamber modules and mechatronic sub-assemblies for FOUP cleaning systems operating in cleanroom class 1 µm environments per SEMI standards.
- **Optothermal Re-Engineering:** Re-engineered IR quartz heater mounting brackets, reducing thermal shadowing by 80% while achieving a 3x weight reduction; verified thermal uniformity via FLIR thermography.
- **PLM & Change Management:** Maintained complex BOMs, authored Engineering Change Notices (ECN), and drafted test reports in Agile PLM and Autodesk Inventor.

Mechanical Design & System Integration Intern — Brooks Automation (Deutschland) GmbH, Steißlingen, Germany 01/2023 – 06/2023

- **Process Air Unit (PAU) Redesign:** Led the mechanical and thermodynamic redesign of the Process Air Unit for EUV Pod Cleaners from initial concept through physical prototype validation.
- **Energy & Footprint Reduction:** Reduced electrical heating power from 48 kW to 18 kW (~50% energy cut), decreased installation volume by 34%, lowered pressure drop by 90% (500 to 50 mbar), and cut manufacturing costs by 21%.
- **Fluid Handling & Standards:** Selected stainless steel alloys (304/316Ti) and integrated fluid control components (valves, flow meters, 4-nm gas filters) adhering to ASME B31.3 and SEMI S8.

Design & Development Engineer (Tooling & Components) — Ravi Brass Industries, Jamnagar, India 08/2020 – 08/2021

- **Precision Electrical Connectors:** Engineered precision micro-connectors using PTC Creo Parametric per DIN/IEC specifications; decreased production scrap by 20%.

KEY ENGINEERING PROJECTS

Optothermal Quartz Heater Module for Vacuum Polymer Processing — Master's Thesis (Grade: 1.3)

Brooks Automation / EAH Jena

- Executed coupled structural and thermal FEA simulations in ANSYS Workbench to optimize radiant heat transfer within FOUP cleaning vacuum chambers.
- Achieved temperature uniformity of ±2 °C under high vacuum; experimentally validated thermal fields using FLIR infrared thermography and optical metrology.

Thermodynamic Redesign of Process Air Unit (PAU) Brooks Automation — Industry Project

- Conducted fluid flow and thermodynamic optimization of high-temperature cleanroom process air heaters for semiconductor lithography tools.
- Decreased power consumption from 48 kW to 18 kW (~50% energy savings), reduced footprint by 34%, and lowered manufacturing costs by 21%.

Modular Custom Machine Platform Standard & DFM/DFA NeSt GmbH Sondermaschinenbau — Series Production

- Standardized modular structural frames, drive housings, and guide mechanisms across 8+ automated production lines.
- Integrated comprehensive DIN EN ISO 1101 GD&T specifications into 200 production drawings per month within SAGE ERP.

EDUCATION

Master of Science (M.Sc.) in Scientific Instrumentation Final Grade: 1.8 (ECTS A) — 10/2021 – 03/2025

Ernst-Abbe-Hochschule Jena, Germany

- **Master's Thesis (Grade: 1.3):** "*Comparative Study and Thermomechanical FEA Simulation of Radiant Heater for High-Vacuum Polymer Processing*" (conducted with Brooks Automation GmbH).
- **Key Focus:** Optical Instrumentation, Precision Engineering, Sensor Technologies, Structural & Thermal FEA (ANSYS), Advanced Materials.

Bachelor of Engineering (B.E.) in Mechanical Engineering Final Grade: 1.9 (CGPA 8.14 / 10.00) — 08/2017 – 08/2020

Gujarat Technological University (ADIT), India

Diploma in Mechanical Engineering Final Grade: 1.2 (CGPA 9.50 / 10.00) — 08/2014 – 06/2017

Government Polytechnic Jamnagar, India

CERTIFICATIONS & TRAINING

ELEATION: ANSYS Professional Certification (FEA Structural & Thermal Simulation)	Certified (325/325)
Siemens NX: Basic to NC Programming & Parametric Surface Modeling	Certified
Semiconductor Innovation: Cleanroom Protocols (Class 1 µm), SEMI S8 & Vacuum Systems	Certified

TECHNICAL SKILLS & COMPETENCIES

CAD Systems:	**SolidWorks**, Autodesk Inventor 2024 (Expert), PTC Creo Parametric, Siemens NX (Sheet metal unfolding, complex surfacing, DIN/ISO 2D drawings)
FEA & Simulation:	**ANSYS Workbench** (Structural & Thermal analysis), GD&T (**DIN EN ISO 1101**), CFD fundamentals, MATLAB & Simulink
Engineering Methods:	DFM/DFA (Design for Manufacturing & Assembly), FMEA risk mitigation, Prototype assembly, Supplier coordination, Engineering changes (ECN)
PDM / PLM & ERP:	Autodesk Vault, Agile PLM, SAGE ERP (Revision control, structured BOM creation and lifecycle management)
Metrology & Testing:	FLIR Infrared Thermography, Optical sensor metrology, Vacuum leak detection, Pressure test benches
Standards & Codes:	Machinery Directive 2006/42/EC, Semiconductor cleanroom standards (SEMI S8 / S2), Process piping (ASME B31.3), DIN/ISO norms
Languages:	**English (C1 — Full Professional / Fluent)**, **German (B2 — Professional Working Proficiency)**, Hindi (Native)